

Optical Character Recognition of Low-Resource Languages

Lightweight OCR for North Sámi Heritage Digitization

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Bachelor Thesis 2026 • IISA 2026 full paper • SCAI 2026 abstract

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Outline

1 Motivation & Research Question

2 Background

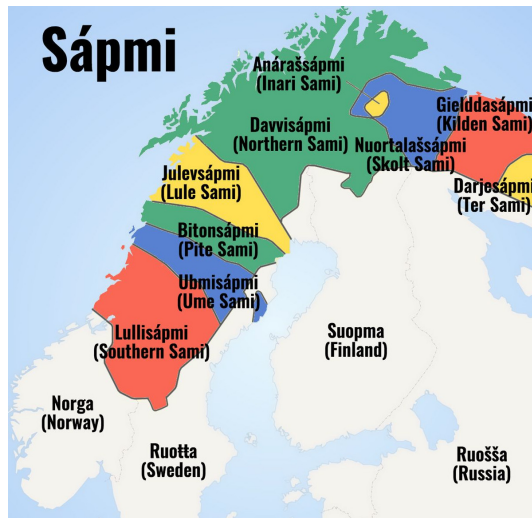
3 What was built

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Why this matters

- North Sámi is an indigenous Uralic language with ~25,000 speakers in Northern Europe
- classified by UNESCO as **definitely endangered**
- Heritage digitization is crucial for perserving the language and culture
- My personal connection: low-resource Silesian language in southern Poland

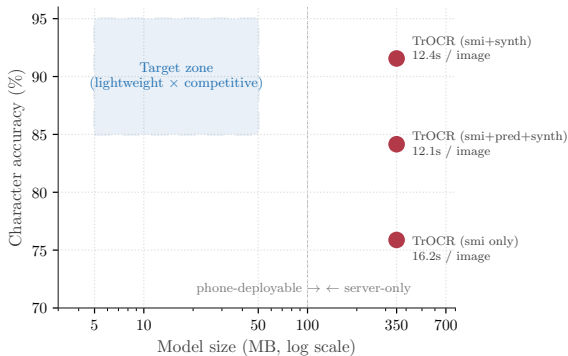


The gap

State of the art: TrOCR (National Library of Norway)

- 91.45% character accuracy
- 350 MB on disk
- 12+ s / image on CPU

Not well fitted for low-budget institutions or individuals in low-resource settings such edge, mobile devices.



Accuracy vs. size/latency.

Research question & contributions

Research question

Can a lightweight OCR architecture approach Transformer-level accuracy for North Sámi while enabling practical deployment on modest hardware?

Six contributions (listed in IISA paper):

- 1 Lightweight CNN-CTC architecture for North Sámi OCR
- 2 Systematic benchmark vs. 7 TrOCR + 3 pretrained-CTC variants
- 3 Sustainable-AI footprint: 380× faster, 15× smaller
- 4 Conjunction-based sentence splitting (proof of concept)
- 5 End-to-end OCR→MT pipeline with TartuNLP
- 6 Applications in 21st-century education, NLP & robotics

What makes North Sámi hard for OCR

1 Seven special diacritics

á, č, đ, ŋ, š, ȥ, ž

2 Agglutinative morphology

Hundreds of inflected forms per verb,
large type-token ratio → model cannot
memorise word shapes.

3 Consonant gradation

giella (“language”) vs. *giela* (“of
language”) — a single dropped “l”
changes the meaning.

okta lea dáhpáhus go bodii dárrolaš njuorjovuopmegierragii

It once happened that a Norwegian settler came to the uplands

Á/á áhku (grandmother)	Č/č čáhci (water)	Đ/đ odda (new)	Ŋ/ŋ eaggas (english)	Š/š šaldi (bridge)	Ȧ/ȥ mákkas (possible)	Ž/ž ieža (other)
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Seven diacritical characters distinguish Sámi from standard Latin alphabet

Seven characters foreign to standard Latin script.

Two architectural families

CNN-CTC (CRNN family)

- CNN → column features
- Reads image left to right
- No alignment supervision needed
- Small, fast

TrOCR (Transformer)

- ViT encoder + autoregressive decoder
- Image patches → self-attention
- Token-by-token generation
- Large, accurate, slow

Same task, very different inductive biases and deployment profiles.

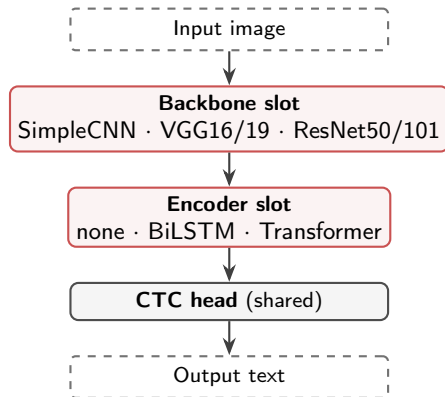
A modular OCR pipeline

Design space: 5 backbones $\times$ 3 encoders, shared CTC head

- Backbones: SimpleCNN, VGG16/19, ResNet50/101
- Encoders: none, BiLSTM, Transformer
- $\rightarrow$ 15 configurations

Trained only 4 due to compute budget (V100, single GPU).

Architecture deliberately extensible, not incomplete.



Three swappable slots, queue training.

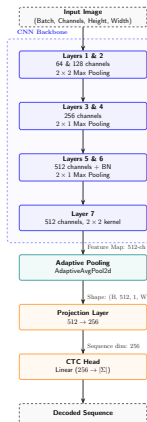
The headline model: CNN-CTC (ctc_simple)

- 7-layer SimpleCNN trained from scratch
- No encoder (direct CNN $\rightarrow$ CTC)
- $\sim 6\text{M}$ params, 23 MB on disk

Asymmetric pooling to reduce height while preserving horizontal resolution

- 2×2 early layers (collapse both axes)
- 2×1 later layers (preserve horizontal resolution)
- Adaptive pool to height 1 $\rightarrow$ column sequence

Preserves character ordering that CTC depends on.



Experimental setup

Train Språkbanken synthetic Sámi OCR dataset (National Library of Norway), ~307k line images (276k training + 31k hyperparameter tuning), 32×800px.

Test 1,048 samples

- 1,000 in-domain synthetic lines (Språkbanken validation split)
- 48 out-of-domain parallel Northern Sámi–English sentences from Turi's 1910 "An Account of Sámi"

Metrics CER, WER, exact match, normalised accuracy (raw UTF-8)

Hardware Nvidia V100 32GB for training and **CPU-only inference benchmark** to simulate resource-constrained deployment.

Main benchmark

Model	Char. Acc. % ↑	Word Acc. % ↑	sec / img	Size (MB)
trocr_smi_synth	91.45	71.84	12.41	~350
ctc_simple	87.76	62.65	0.033	23
trocr_smi_pred_synth	84.15	54.24	12.08	~350
trocr_smi	75.88	36.50	16.23	~350

Char. Acc. = 100 - CER Word Acc. = 100 - WER

Results for ctc_simple:

87.76% char. acc. (within 3.69 pp of best TrOCR), 380× faster, 15× smaller.

Consequently: 1 M-page corpus takes <10 hours on CPU vs. ~144 days for TrOCR.

12 / 19

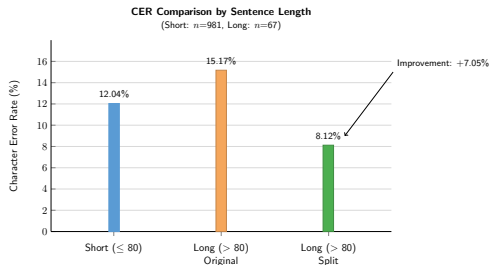
Conjunction-based sentence splitting

Long sentences are over the fixed 800 px input width.

Split at North Sámi conjunctions (*ja*, *muhto*, *dahje*, *go*, *ahte*, *jos*), OCR each segment, rejoin.

- Long-sentence CER: 15.17% → 8.12% (−7.05 pp)
- Below short-sentence baseline (12.04%)
- $p = 0.000009$, Cohen's $d = 1.35$
- 95% of split samples improved, none degraded

Oracle upper bound: splitting operates on ground-truth strings, the cut images are re-rendered without noise. After deployment, much of the gain would be lost.



End-to-end OCR → MT pipeline

Predicted text from `ctc_simple` fed to TartuNLP's North Sámi → English API.

Metric	Score
BLEU	10.40
chrF	33.85
TER	76.30%

Why moderate? Character errors land on **morphological suffixes** that carry grammatical role – a single dropped character flips the meaning of a whole clause.

Good for *indexing / content discovery*, not full translation.

Example (entry 998, CER 1.2%):

Ja áldoeallu lea váralaš: dan galggalii sáhhtit nu siivvut reainnidit go lea mátolaš.

GT (sámi): *ja áldoeallu lea váralaš dan galggalii sáhhtit nu **siivvut** reainnidit go lea mátolaš*

Pred (sámi): ...sáhhtit nu **siivut** reainnidit ... [one “v” dropped]

GT (en): “and it is a **dangerous time for the cow herd**, they must be **herded** as carefully as possible.”

Pred (en): “and **the livestock is dangerous**, it should be able to be **removed** as safely as possible.”

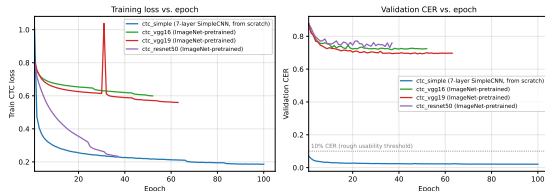
Negative result: ImageNet pretraining failed

Three pretrained backbones – VGG16, VGG19, ResNet50 – all >69% CER.

- VGG pair: *underfits* (train loss barely moves)
- ResNet50: *overfits* (train collapses, val stuck)
- Same head and projection layer as `ctc_simple`

Hypothesis: ImageNet's object-level features misalign with character-stroke topology.

Couldn't try again due to compute budget.



Were goals achieved? - *partially*

Aspect	Answer	Why
Deployment	Yes	23 MB, 0.033 s/image, <100 ms latency, deployable on edge devices
Accuracy	Yes, but...	3.69 pp behind best TrOCR, out-of-domain is better on normalised acc.
Validation	Open	Synthetic-only evaluation, no real scans yet, lack of statistical power

Applications & future work

Applications

- Mobile / edge OCR for community digitalization
- Large (milions) scanned documents indexing
- Joint OCR + reconstruction of occluded text (Figure)

Future work

- Real scanned corpus
- Giellatekno morphological post-correction
- Layout analysis for full pages
- Transfer to other Sámi varieties (Inari, Skolt, Lule)



*Damaged-text reconstruction proof of concept:
OCR + diffusion inpainting on an artificially occluded
line.*

Conclusion

Can a lightweight OCR architecture approach Transformer-level accuracy for North Sámi while enabling practical deployment on modest hardware?

A 23 MB, edge-ready CNN-CTC OCR
within 3.69 pp CER of state of the art,
380× faster, with stable historical-text behaviour

- making the deployment easier for communities for which the technology should serve.

Thank you

Questions?

`jamus23@student.sdu.dk`
`github.com/magwrap/north-sami-ocr`
`huggingface.co/magwrap/cnn-ctc-ocr-sme`

Additional slides

Other models: Tesseract / Transkribus comparison

- Transkribus achieved the lowest CER for most languages (overall CER of 0.61%).
- TrOCR obtained the best word error rate (overall WER of 2.96% vs. Transkribus's 3.19%) and the best special-character F1 score (96.97% vs. Transkribus's 96.03%).
- Tesseract trailed both (CER 0.89%, WER 4.65%, F1 93.81%).

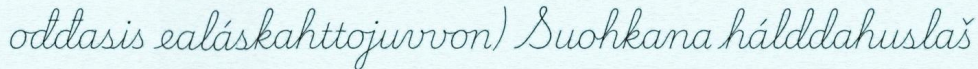
Per-character diacritic accuracy

Character	Count	ctc_simple	trocr_smi_synth
á / Á	1473	90.1%	95.9%
č / Č	129	83.7%	81.4%
ď / Ď	191	79.1%	87.4%
ŋ / Ŋ	26	100.0%	84.6%
š / Š	262	87.8%	93.1%
ť / Ť	6	50.0%	50.0%
ž / Ž	35	80.0%	85.7%
Overall	2122	88.3%	93.4%

- TrOCR ahead overall, but ctc_simple **perfect on ŋ**, ahead on č
- Both fail on ě (rarity: only 6 occurrences, visually $\approx$ t)
- Common errors: á→a, ď→d (stroke missed)

Examples of failed predictions for CNN-CTC

Diacritical error (CER = 8.3%)



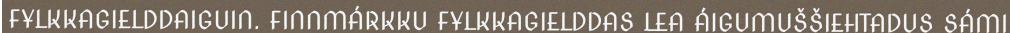
odd̄asis ealáškahttojuvvon) Suohkana hálddahuslaš

GT: odd̄asis ealáškahttojuvvon) Suohkana hálddahuslaš

Pred: oddasis ealáškahttojuvvon) Duohkama hálddahuslaš

– classic đ→d stroke miss.

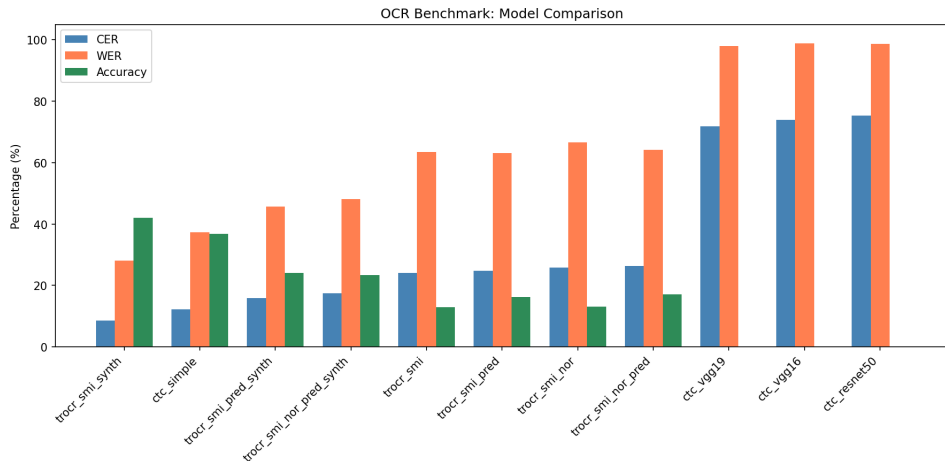
Uppercase error (CER = 29.6%)



FYLKKAGIELDDAIGUIN. FINNMÁRKKU FYLKKAGIELDDAS LEA ÁIGUMUŠŠIEHTADUS SÁMI

All-uppercase lines rare in training distribution → digit substitutions / case flips.

Full 11-model benchmark



All 11 models on the 1,048-sample test set.